

Exchange rate and Production network

Macro Internal Seminar

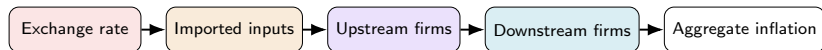
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Motivation

A depreciation raises the cost of imported inputs. But firms don't just buy imports directly — they buy from *other domestic firms*, who themselves buy imports. A single exchange-rate shock therefore travels through the production network before it ever reaches the consumer: **the same shock can be small or large, depending on where in the network it lands.**



Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): get this propagation right — but **closed economy**, no exchange rate

Open-economy NK (Galí–Monacelli 2005): get the exchange rate right — but **no network**, one representative firm

This paper: both together — and asks the question that requires having both

Does accounting for production networks change the optimal exchange-rate regime?

Why Production Networks Matter

Intermediate inputs dominate production. In most economies, firms spend more buying from *other domestic firms* than they spend on labor — a cost shock to any one sector mechanically becomes a cost shock to everyone downstream of it

Imported intermediates are the norm, not the exception. Manufacturing sectors routinely source 15–30% of inputs from abroad (Chile: up to 19.5%, Manufacturing) — so an exchange-rate move is, from the firm's perspective, indistinguishable from a domestic cost shock

Recent shocks propagated through exactly this channel. COVID-19, the Ukraine war, and the semiconductor shortage each started as a shock to one link and were amplified — or dampened — by where that link sat in the network

Acemoglu, Carvalho, Ozdaglar & Tahbaz-Salehi (2012); Baqaee & Farhi (2019); La'O & Tahbaz-Salehi (2022)

Literature

Network side (closed economy):

Rubbo (2024) — Calvo network $\Rightarrow$ unique DC index (1 cost-push channel)
La'O & Tahbaz-Salehi (2022) — optimal MP in production networks
Pasten, Schoenle & Weber (2020) — sectoral rigidity heterogeneity drives MP transmission

Open-economy side (no network):

Galí & Monacelli (2005) — 1-sector SOE: domestic-inflation targeting $\approx$ optimal, peg dominated (ToT channel)
Fanelli & Straub (2021) — FX intervention as a *second* instrument, separate from i_t
Schmitt-Grohé & Uribe (2003) — debt-elastic premium to stationarize NFA; flags 1st-order welfare risk
Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019) — pass-through, dominant currency, IO spillovers
Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023) — frontier: managed float/fear-of-floating *optimal*, not behavioral (no network)

Open economy with network (closest to this paper):

Gnoco, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025) — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice¹
Qiu, Wang, Xu & Zanetti (2026, JME) — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice — their own conclusion lists relaxing financial autarky as their top open extension

Gap: network + open economy + FX **regime choice** together — no one has this. Same ranking as Galí–Monacelli but a **different channel** (risk-premium, not ToT); Managed's dominance echoes Fanelli–Straub & Itskhoki–Mukhin, but from network persistence.

→ paper-by-paper comparison to the literature: Appendix

¹Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo $\rightarrow$ SOE: imported inputs Ω^F , UIP, debt-elastic premium

Ask: optimal FX regime? DC index survive with 2 cost-push channels?

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Not just Galí-Monacelli with sectors: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new

Calibrate: Chile's real I-O table (Banco Central), also Korea, Czechia & a stylized baseline; solve **nonlinear** model directly

Households

Small open economy: home takes foreign price P_t^{F*} , rate i^* , and demand D_t^* as given.²

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize CRRA-consumption/disutility-of-labor utility, subject to a budget constraint with a domestic bond B_t (nets to zero in equilibrium) and a foreign bond B_t^* :

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1})B_{t-1} + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)] S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

² Full equations & notation for every block in this section: Appendix.

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition; CRS makes marginal cost common across firms:

$$Y_{ift} = A_{it} L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H} M_{ift}^{\Omega_i^F}$$
$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) and S_t (import cost); with **Calvo pricing**, only a fraction δ_i resets each period — mechanics & Phillips curve: next slide.

Production Network

Stacking each firm's input shares $\Omega_{ij}^H, \Omega_i^F$ across sectors gives the **input-output network** $\Omega^H \in \mathbb{R}^{N \times N}$ and import-exposure vector $\omega_F \in \mathbb{R}^N$.

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \dots$$

applied to consumption shares β^H (Households) it gives each sector's **Domar weight** (total GDP sales share); applied to ω_F , its total foreign exposure:

$$\lambda^\top = (\beta^H)^\top (I - \Omega^H)^{-1}, \quad \mathcal{M} = (I - \Omega^H)^{-1} \omega_F$$

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\pi_t = \rho(I - \mathcal{V})\mathbb{E}_t\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \Gamma\Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator

$\tilde{A} \equiv \Delta(I - \Omega^H\Delta)^{-1}$, applied to labor share α vs. import share ω_F :

$$\mathcal{B} = \frac{\tilde{A}\alpha}{\kappa}, \quad \Gamma = \frac{\tilde{A}\omega_F}{\kappa}$$

Only relative shocks matter: $\mathcal{V}\mathbf{1} = \mathbf{0}$ — uniform TFP shocks create no cost-push; only sector-specific gaps $\chi_t = (I - \Omega^H)^{-1} \log \mathbf{A}_t - \mathbf{p}_{t-1}$ do.

where:

$\pi_t, \tilde{y}_t$: infl., gap

χ_t : TFP cost-push

Δe_t : exch. rate chg. κ : GE scalar

$\mathcal{B}, \Gamma$: PC slope, FX pass-thru

$\mathcal{V}$: TFP cost-push matrix

$\tilde{A}$: rigidity-adj. Leontief

Foreign Sector

P_t^{F*} , i^* , D_t^* are exogenous AR(1) processes. Law of one price:

$$P_t^{FH} = S_t P_t^{F*}$$

UIP, with a debt-elastic risk premium closing the model (Schmitt-Grohé–Uribe) and RP_t the risk-premium shock studied below:

$$i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance; exports are **sector-specific**, each responding to its own relative price (baseline model — not a single aggregate EX_t allocated by consumption shares):

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \quad EX_{it} = \kappa_{EX,i} D_t^* \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

$\kappa_{EX,i}$ is calibrated so each sector's steady-state export share matches Chile's real data (Resource-heavy: 0.602/0.180/0.036) — not the Services-heavy household shares β_i^H that a single aggregate EX_t would imply.

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \rightarrow MC_{it} \rightarrow \pi_{it} \quad (\text{cost-push}) \quad \hat{s}_t \rightarrow EX_{it} \rightarrow \tilde{y}_t \rightarrow \pi_{it} \quad (\text{demand})$$

Shocks (7 total, all AR(1), std. dev. 0.01): 3 sector TFP $\varepsilon_{1,2,3}^A$ ($\rho=0.90$) · import price $\varepsilon^{P^{F*}}$ ($\rho=0.85$, into P_t^{F*}) · foreign demand ε^{D^*} ($\rho=0.80$) · export price/ToT ε^{PX} ($\rho=0.85$) · risk premium ε^{RP} ($\rho=0.80$, into RP_t above) — decomposed by contribution to welfare loss in Results.

Monetary Policy Regimes

Three exchange-rate regimes are compared. All target the **DC index** π_t^{DC} , **not CPI inflation**: Rubbo (2024) shows CPI targeting does *not* achieve divine coincidence even in the closed economy — π_t^{DC} is the closed-economy-optimal benchmark, so it is the natural rule to test for survival once the economy is opened.

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Everything above collapses to four blocks — this is what is actually coded up and simulated in Dynare.

Phillips curve (vector, network-amplified, two cost-push channels):

$$\pi_t = \rho(I - \mathcal{V})\mathbb{E}_t\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \Gamma\Delta e_t$$

IS curve (aggregate demand; open economy adds the same relative-price term as the cost-push channel above):

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t\pi_t^C - r_t^{\text{nat}}) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H)$$

Foreign sector (UIP closes the FX block; NFA is the only state carried across regimes):

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t, \quad \hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C}\hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}$$

Exports are sector-specific, $EX_{it} = \kappa_{EX,i} D_t^* (P_{it}/P_t^{FH})^{-\theta^*}$ (baseline; $\kappa_{EX,i}$ calibrated to Chile's real export shares).

Policy rule (regime-dependent; the only block that changes across Float/Peg/Managed — same rules as previous slide):

$$\hat{i}_t = \begin{cases} \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t & \text{Managed} \end{cases}$$

where:

Welfare Loss Function

$$\mathcal{W} = \frac{1}{2} \left[\underbrace{(\gamma + \varphi) \text{Var}(\tilde{y}_t)}_{\text{output gap}} + \underbrace{\sum_i \kappa_i \text{Var}(\pi_{it})}_{\text{price dispersion}} \right], \quad \kappa_i = \lambda_{D,i} \varepsilon \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

Calibration: Chile Data

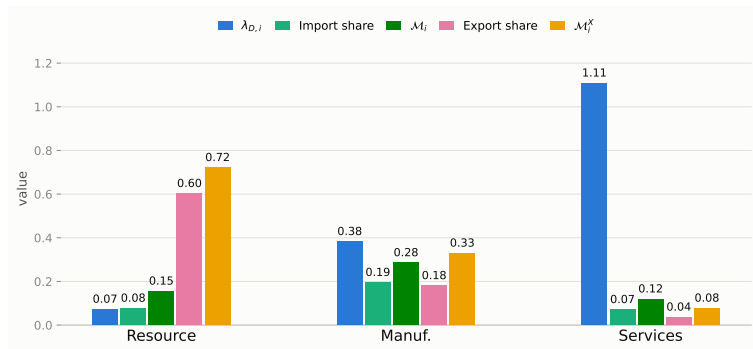
Parameter	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604	Data: BCCh IO table
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744	
Export share _{1,2,3}	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); Nakamura–Steinsson (2008)
ψ	0.020	Key mechanism: risk-premium/UIP elasticity & FX-regime rules (see ψ -sensitivity, appendix)
ϕ_π	1.50	
ϕ_y	0.50	
ϕ_s	0.30	

Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Standard literature values (not paper-specific; sources in Appendix): $\beta=0.99$, $\gamma=1.00$, $\varphi=2.00$, $\eta=1.50$, $\varepsilon=8.00$, $\theta^*=2.00$, $\beta^F, \text{tot}=0.10$.

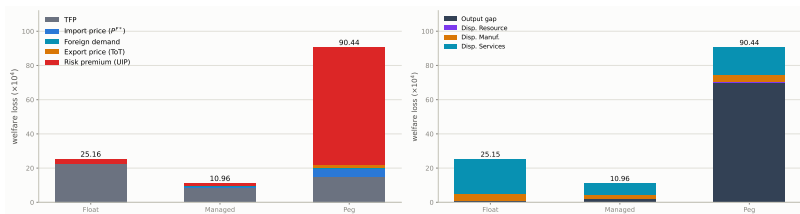
Network Properties: Import & Export Exposure



The production network amplifies both import and export exposure well beyond each sector's direct (raw) share — e.g. import exposure roughly doubles for every sector once indirect, network-propagated exposure is counted, not just each sector's own imports.

$\lambda_{D,i} = [(\beta^H)^\top (I - \Omega^H)^{-1}]_i$ (Domar weight), $\mathcal{M}_i = [(I - \Omega^H)^{-1} \omega_F]_i$ and $\mathcal{M}_i^X = [(I - \Omega^H)^{-1} \omega_X]_i$ (import/export centrality) — full notation: Appendix.

Welfare Ranking: Managed Float Dominates



Managed float dominates both corners: Float 25.15, Managed 10.96, Peg 90.45 ($\times 10^{-4}$, real Chile calibration) — Managed < Float $\ll$ Peg, Peg no longer competitive

By shock (left): risk-premium/UIP dominates Peg (76% of its total). **By gap vs. dispersion (right):** Peg's loss is $\approx 77\%$ output gap (70.01 of 90.45, zero monetary autonomy); Float's and Managed's are almost entirely price dispersion, concentrated in Services — full decomposition in appendix

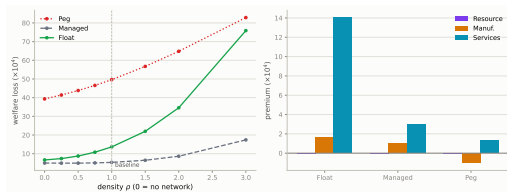
This ranking is the headline result of the paper. Everything that follows asks: where does it come from, and does the network matter for it?

Isolating the Network Channel

The causal experiment: Network OFF ($\Omega^H \equiv 0$) vs. Network ON (real Chile Ω^H) — same shocks, same regimes, only the domestic input-output structure changes.

Total welfare loss ($\times 10^4$)	Float	Managed	Peg
Network OFF	8.67	5.82	33.66
Network ON (baseline)	25.15	10.96	90.45
% increase from the network	+190%	+88%	+169%

The network substantially increases welfare losses under Float and Peg but barely affects Managed — and Managed stays the best regime either way.



Continuous density dial ρ (stylized network) confirms it: losses rise **monotonically** with density, and the Peg/Float gap **widens** as the network gets denser.

Mechanism: How the Network Amplifies Shocks

Output-gap variance, **Network ON** vs. **Network OFF** (real Chile calibration) — full IRFs:

Appendix:

Regime	Import-price shock	Risk-premium/UIP shock
	$\text{Var}(\tilde{y}_t)$ ratio, ON/OFF	$\text{Var}(\tilde{y}_t)$ ratio, ON/OFF
Float	5.4×	8.5×
Managed	6.0×	1.1× (<i>essentially none</i>)
Peg	11.1×	4.8×

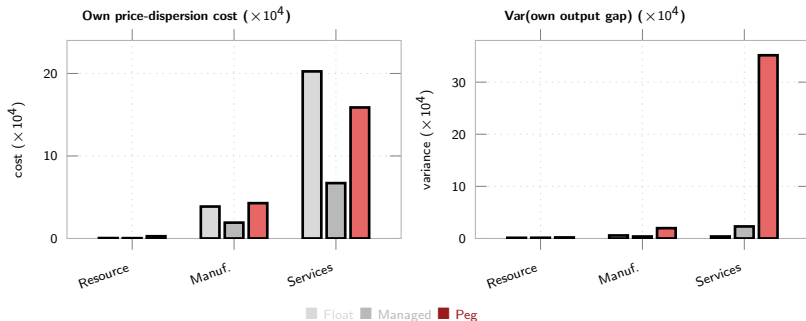
Amplification is regime-and-shock-specific, not a generic network multiplier. Import-price shocks hit **Peg** hardest (no FX buffer to absorb pass-through); risk-premium shocks hit **Float** hardest ($\hat{s}_t$ free to move transmits the shock straight through the network) — **Managed** blocks amplification of the risk-premium shock almost entirely, but not the import-price shock

DC inflation variance is unchanged or slightly lower with the network on, for *both* shocks in every regime — this isn't new cost-push, it's a redirection of the existing shock into output

Takeaway: which shock dominates a regime's loss (Welfare Ranking, above) depends on which margin that regime leaves free to adjust

Network Exposure & Sector Welfare Preferences

Different sectors prefer different exchange-rate regimes, in both price and quantity. A representative-sector model cannot generate this: it only exists because sectors differ in Calvo stickiness $\hat{\delta}_i$ and network centrality $\lambda_{D,i}$, so the same shock hits their welfare weight $\kappa_i = \lambda_{D,i} \varepsilon \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ and their output-gap exposure differently.



Left (price dispersion): Managed dominates every sector. Peg costliest for Resource & Manuf.; for Services, **Float is costliest** ($20.25 > \text{Peg's } 15.88$) — near-unit-root S_t under Float feeds the stickiest sector's marginal cost for a long time. Aggregate ranking is set by Services, which carries most of the economy's welfare weight

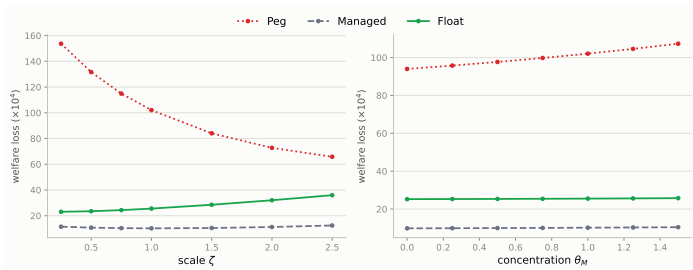
Right (output-gap variance, descriptive — the loss function only weights the aggregate gap): opposite pattern — **Peg is the outlier**, concentrated almost entirely in Services

Openness & Import Concentration

Reparametrize ω_F only, on the REAL Chile calibration ($\alpha_i = 1 - \Omega_i^H - \omega_{F,i}$ absorbs it, holding Ω^H fixed):

$$\omega_{F,i}(\zeta) = \zeta \omega_{F,i}^{\text{base}} \quad \omega_{F,i}(\theta_M) = \bar{\omega}_F + \theta_M (\omega_{F,i}^{\text{base}} - \bar{\omega}_F)$$

$\zeta = \theta_M = 1$: baseline. ζ : uniform rescaling. θ_M : mean-preserving redistribution ($\theta_M=0$: uniform exposure).



Managed < Float < Peg robust throughout (aggregate-export baseline, predates the sector-specific-export switch, not yet rebuilt); gap moves in **opposite directions**:

ζ (openness): Peg $\downarrow$ (153.7 $\rightarrow$ 65.8), Float $\uparrow$ (23.1 $\rightarrow$ 36.0) — **converge**

θ_M (concentration): Peg $\uparrow$ faster (94.0 $\rightarrow$ 107.3) than Float (25.2 $\rightarrow$ 25.8) — **gap diverges**, via Peg rising, not Float falling

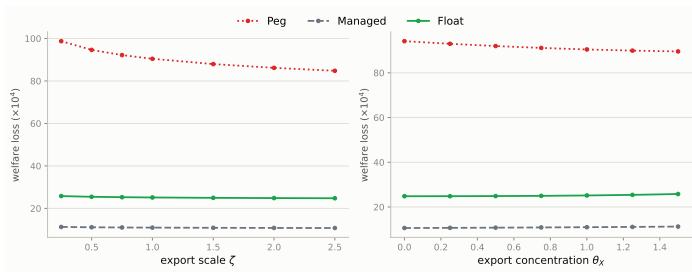
Mechanism: $\alpha_i + \Omega_i^H + \omega_{F,i} = 1$ links labor- and import-cost channels; Peg's risk-premium/output-gap channel is far more sensitive to reallocation toward Services than Float's price-dispersion channel

Export Openness & Export Concentration

Same construction, applied to the sector-export model's $\kappa_{EX,i}$ (real Chile calibration; $\zeta = \theta_X = 1$ recovers 25.15/10.96/90.45 exactly):

$$\kappa_{EX,i}(\zeta) = \zeta \kappa_{EX,i}^{\text{base}} \quad \kappa_{EX,i}(\theta_X) = \bar{\kappa}_{EX} + \theta_X (\kappa_{EX,i}^{\text{base}} - \bar{\kappa}_{EX})$$

$\zeta = \theta_X = 1$: baseline. ζ : uniform export-openness scaling. θ_X : mean-preserving redistribution ($\theta_X=0$: uniform export exposure).



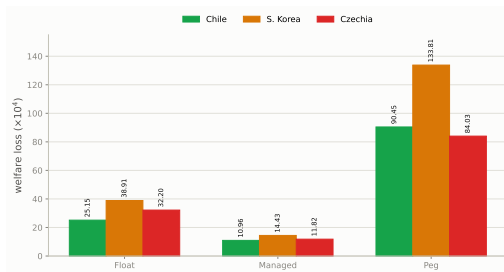
Managed < Float < Peg robust throughout. Unlike import ζ (opposite directions — $\omega_{F,i}$ trades off against α_i), export ζ moves **all three regimes together**:

ζ (openness): **all regimes improve** as trade opens — Peg most (98.7 $\rightarrow$ 84.8), since $\kappa_{EX,i}$ only enters export demand, not the cost side

θ_X (concentration): **opposite directions, as with import concentration** — Peg $\downarrow$ (94.1 $\rightarrow$ 89.6), Float/Managed $\uparrow$ slightly, as exports concentrate toward Resource (the flexible-price sector)

Mechanism: routing exports toward Resource's low κ_1 helps Peg (flexible-priced sector absorbs shocks better); raises Float/Managed's dispersion cost slightly instead

Robustness: South Korea & Czechia



The ranking is robust: Managed < Float $\ll$ Peg in **every** calibration. UIP dominates Peg's loss throughout (76% Chile; 72%/65% Korea/Czechia). Also robust to a 2nd-order solution and to the import/export openness & concentration sweeps above.

Why Korea is uniformly higher (Float and Peg): denser network (Ω^H row sums 0.50/0.50/0.39 vs. Chile's 0.42/0.45/0.33) *and* higher import exposure, esp. Manufacturing ($\Omega_2^F=0.24$ vs. 0.195, electronics/machinery supply chains) $\Rightarrow$ import centrality $\mathcal{M}_i$ higher in every sector (Manuf. 0.41 vs. 0.29) $\Rightarrow$ bigger Γ (Float's price-dispersion channel) *and* more network amplification of the output gap (Peg's channel) — same two levers as "Isolating the Network Channel," just further along the dial.

Caveat: Chile uses the sector-specific-export baseline; Korea/Czechia still use the earlier aggregate-export (β^H -allocated) spec —

illustrative of network/calibration robustness, not a fully apples-to-apples export comparison.

Conclusion

Managed float optimal: 10.96 vs. Float 25.15, Peg 90.45 ($\times 10^4$, Chile calibration, sector-specific-export baseline)

Driver: risk-premium/UIP dominates Peg (76%); TFP persistence dominates Float & Managed

Network channel isolated (network OFF vs. ON, same baseline): losses rise +190% Float, +88% Managed, +169% Peg — **Peg's rise is almost entirely an output-gap story** (variance amplifies $11\times$ for this shock; price dispersion barely moves)

Ranking robust to South Korea & Czechia calibrations and to a genuine 2nd-order (nonlinear) solution — those two and the import/export openness sweeps still reference the earlier aggregate-export specification, pending rebuild

Vs. literature: same ranking as Galí–Monacelli, different channel (risk-premium, not ToT)

Next: many-sector OECD TiVA calibration; rebuild Korea/Czechia & the $\phi_s/2\text{nd-order}/\psi$ robustness checks on the sector-specific-export baseline

Appendix: Chile Calibration — Sector Concordance & Sources

Resource = mining/copper (Chile's export engine, 60% of own output exported) → Manuf. (mid-chain) → Services (largest value added, sticky)

$\alpha_i, \Omega_{ij}^H, \Omega_i^F, \beta_i^H$: **data** (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. 2023)

$\delta_1, \delta_2, \delta_3$: **literature** — euro-area price-change frequencies (Dhyne et al. 2005/ECB IPN) → quarterly reset prob.; cross-checked vs. Nakamura & Steinsson (2008) US PPI

$\beta, \gamma, \varphi, \eta, \varepsilon, \theta^*, \psi, \phi_\pi, \phi_\gamma, \phi_s$: literature-standard

Sector concordance (Banco Central de Chile, CdeR 2018, 12 activities → our 3): **Resource** = {1 Agriculture, forestry & fishing, 2 Mining}; **Manufacturing** = {3 Manufacturing industry, 4 Electricity, gas, water & waste management, 5 Construction}; **Services** = {6 Trade, hotels & restaurants, 7 Transport, communications & information services, 8 Financial intermediation, 9 Real estate & housing services, 10 Business services, 11 Personal services, 12 Public administration}.

Import & export data exist at the sector level from the same tables: imports = imported intermediate input use per buying sector (sheet 21, underlies Ω_i^F); exports = exports of each sector's own output (sheet 20, underlies the export-share figures on the Calibration slide).

Appendix: Is the Ranking Robust to a Genuine 2nd-Order Solution?

Why this was a caveat: Rubbo's LQ welfare formula is 2nd-order in utility, but evaluated on *1st-order* (certainty-equivalent) paths — valid only if $\mathbb{E}[\tilde{y}_t] = \mathbb{E}[\pi_{it}] = 0$ exactly. The **debt-elastic UIP wedge** and **near-unit-root NFA** can break that.

Note: run on the earlier aggregate-export specification, not yet re-verified on the sector-specific-export baseline; the mechanism is unrelated to the export side, so the qualitative conclusion should carry over.

Method: model already coded nonlinear (levels) in Dynare. Solve **order 2 with pruning**, simulate 260k periods, compute $\mathbb{E}[X^2]$ directly. Control for simulation noise with an identical-seed order=1 run.

Welfare loss ($\times 10^4$)	Float	Managed	Peg
1st-order (analytic, headline)	25.47	10.17	102.05
1st-order (simulated, same seed)	25.58	10.20	102.39
2nd-order (pruned, simulated)	26.09	10.26	103.13
Net 2nd-order effect	+2.0%	+0.6%	+0.7%

Ranking survives: Managed < Float < Peg at 2nd order, correction $\leq 2\%$ everywhere

Risk-adjusted means are real but small: Peg's output gap sits $\approx -45\text{bp}$ on average (vs. $-2.5/-7\text{bp}$ Float/Managed); NFA carries a precautionary buffer under Peg

But squared loss barely notices: $\mathbb{E}[X]^2/\mathbb{E}[X^2] < 0.5\%$ everywhere — almost all the correction is variance amplification, not the risk-adjusted-mean channel

Appendix: Second-Order Welfare
Check (Backup)

Appendix: Is the 75% Risk-Premium Share a ψ Knife-Edge?

Why this was a caveat: the "75% risk-premium" result rests on a debt-elastic UIP closing device (Schmitt-Grohé–Uribe 2003) at $\psi = 0.020$. Known pitfall: ψ near zero can create a near-unit-root NFA blowup that's a linearization artifact — the order-2 check (previous slide) ruled out spurious *nonlinearity*, not whether *this* ψ value does unreasonable work.

Note: grid run on the earlier aggregate-export spec (75.3% baseline share); the sector-specific-export baseline gives 76.1% at $\psi = 0.020$, essentially identical — not yet re-swept, but the closeness of the two baselines suggests this sweep's shape carries over.

Method: scale ψ over $0.25 \times 4 \times$ baseline (0.005–0.080), re-solve all three regimes (nonlinear, Dynare), recompute the risk-premium shock's % share of welfare loss via the exact variance decomposition.

ψ	0.005	0.010	0.020 (base)	0.040	0.080
Peg total loss ($\times 10^4$)	138.17	120.74	102.05	83.69	67.46
Risk-prem. % of Peg's loss	83.0%	80.0%	75.3%	68.1%	57.6%

Not a knife-edge: share declines **smoothly, monotonically** across a $16\times$ range — no blow-up near $\psi \rightarrow 0$ or the calibrated value

Mechanism robust, not the number: even at $4\times$ baseline, risk-premium/UIP still drives a **majority** (57.6%) of Peg's loss

Ranking survives the whole grid: Managed < Float < Peg at every ψ tested
Appendix: ψ -Sensitivity of the
Risk-Premium Share

Appendix: Network vs. No-Network, on the Real Chile Data

Why this is needed: "Isolating the Network Channel" (main deck) uses a **stylized triangular** Ω^H scaled by density dial ρ , not Chile's actual **dense** Ω^H . Here: zero every Ω_{ij}^H (incl. diagonal), let α_i absorb the freed share ($\alpha_i = 1 - \omega_{F,i}$); re-derive the steady state and re-solve all three regimes on the real data.

Total welfare loss ($\times 10^4$)	Float	Managed	Peg
With network (baseline, headline)	25.15	10.96	90.45
No network (counterfactual)	8.67	5.82	33.66
Network premium	+190%	+88%	+169%

Network roughly doubles-to-triples every regime's loss — not a decorative feature, most acutely Float (+190%)

Ranking preserved either way: Managed < Float < Peg with *and* without the network

Network reallocates Peg's loss: without it, output-gap shrinks (70.01 $\rightarrow$ 13.49) while price dispersion is essentially flat (20.43 $\rightarrow$ 20.17)

Appendix: Does Export Concentration Matter, Not Just Export Openness?

Why the baseline uses sector-specific exports: an earlier version allocated aggregate exports EX_t via household shares β_i^H (Services-heavy: 0.744) — *opposite* Chile's real export pattern (Resource-heavy: 0.602/0.180/0.036). The baseline instead uses three sector-specific equations $EX_{i,t} = \kappa_{EX,i} D_t^* P_t^X (P_{i,t}/P_t^{FH})^{-\theta^*}$, $\kappa_{EX,i}$ calibrated (Newton's method) so exports hit the true exporting sector.

Total welfare loss ($\times 10^4$)	Float	Managed	Peg
Deprecated (β^H -alloc., Services-heavy)	25.47	10.17	102.05
Baseline (sector-specific, real data)	25.15	10.96	90.45

Export/ToT shock welfare ($\times 10^4$), deprecated $\rightarrow$ baseline: Float 0.11 $\rightarrow$ 0.17, Managed 0.18 $\rightarrow$ 0.14, Peg 2.16 $\rightarrow$ 1.17

Ranking preserved: $10.96 < 25.15 < 90.45$ once exports are concentrated in the true exporting (Resource) sector

Concentrating exports in the flexible-price sector lowers Peg's ToT cost: Resource ($\delta_1 = 0.90$, low κ_1) vs. Services (stickiest, high κ_3) cuts Peg's ToT welfare cost 46% — this is *why* the baseline switched

Caveat: matching export *shares* exactly shrinks total trade volume $\approx 22\%$ ($0.256 \rightarrow 0.201$); $BSTAR_{ss} = 0$ still holds exactly by construction

Appendix: Why Exports Are
Sector-Specific in the Baseline

Mechanisms: Why Each Regime Costs What It Costs

Regime	Cost channel	What sets the magnitude
Float	S_t 's near-unit-root persistence (NFA state) feeds $\Gamma_i \Delta e_t$ into every sector's MC, even for shocks with no FX content — pure dispersion cost	Persistence of debt-elastic premium (ψ); Γ_i (import exposure); κ_i of the most central/sticky sector (Services: 20.3 of 25.2)
Peg	Zero monetary autonomy: output-gap term explodes (70.0 of 90.4, $\approx 77\%$) on the UIP shock; $\Delta e_t \equiv 0$ <i>relocates</i> , not removes, adjustment into upstream P^H	Absence of ϕ_π, ϕ_y ; real vs. nominal share of the shock; upstream sector's Domar weight
Managed	$\phi_S \Delta \log S_t$ damps $\Gamma_i \Delta e_t$ without fully re-imposing peg's relocation cost — dominates every sector at once	Interior optimal $\phi_S^* \approx 0.20$ (aggregate-export baseline; not yet re-swept): too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ autonomy lost

Two forces, always in tension: Γ -channel favors peg (switches off $\Gamma_i \Delta e_t$); **relocation** channel favors float (peg's fixed e_t forces the same adjustment into upstream prices). Which wins depends on Γ_i vs. κ_i ; managed is the only regime improving on *both* corners for every sector.
Analytical or simulated? Both.

Channels — closed-form: $\Gamma_i = [\tilde{A}\omega_F]_i / \kappa$ (pass-through), $\kappa_i = \lambda_{D,i} \varepsilon^{\frac{1-\delta_i}{\delta_i}}$ (welfare weight),

DC-survival $\phi^\top \mathcal{V} = 0^\top$ and $\phi^\top \Gamma = 0$ — from primitives alone, no model solve.

Magnitudes/ranking — need simulation: $\text{Var}(\pi_{it}), \text{Var}(\tilde{y}_t)$ come from the full linear RE solution (order-1 Dynare) — no closed form once $\phi_S > 0$ or Ω^H dense.

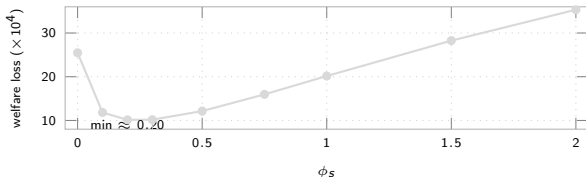
Managed < Float < Peg and $\phi_S^* \approx 0.20$ are simulated, but robust across all four calibrations.

Appendix: Mechanisms — Why
Each Regime Costs What It
Costs

Managed Float: Optimal Stabilization Intensity ϕ_s

Note: this sweep runs on the earlier aggregate-export specification, not yet rebuilt on the sector-specific-export baseline (which gives Peg 90.45, not 102.05, at $\phi_s=0/\text{Float}$) — the interior-optimum *mechanism* (UIP leak-in vs. autonomy loss) is unrelated to the export side, so should carry over, but the exact optimum ϕ_s^* is pending re-verification.

Welfare loss vs. ϕ_s ($\times 10^4$), full 7-shock model, real Chile calibration (aggregate-export specification)



Why interior: too low $\rightarrow$ UIP shock leaks in ($\phi_s=0$ collapses to Float's 25.47); too high $\rightarrow$ loses autonomy

$\phi_s=0 \rightarrow 0.20$: loss 25.47 $\rightarrow$ 10.14. **Interior optimum, but nearly flat near it:** 0.20 beats the calibrated 0.30 by only $\approx 0.3\%$ (was $\approx 13\%$ under the old stylized-network sweep — the Chile calibration's optimum is much less sharply peaked)

Beyond ≈ 0.3 : rises steadily, back above Float (25.47) between $\phi_s=1.0$ (20.16) and 1.5 (28.26)

Appendix: DC Index — Definition and Insulation

Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$ rank-deficient $\Rightarrow$ unique DC index:

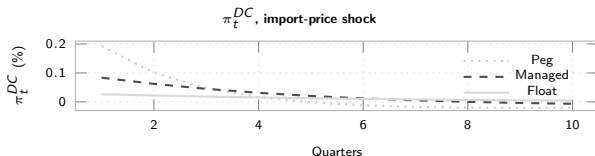
$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}, \quad w_i^{DC} \propto \lambda_{D,i} \frac{1-\delta_i}{\delta_i}$$

Open economy: 2 channels (TFP + FX):

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \text{generically infeasible}$$

Knife-edge test: $\Gamma \propto \mathcal{B}$ would save the DC index. Instead: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ ($\approx 17^\circ$ apart) — close, but $\Gamma_i/\mathcal{B}_i$ shrinks $\approx 2\times$ per sector, not constant. Breakdown is real, not a numerical artifact.

Fix: import nodes flexible ($\hat{\delta}_M=1 \Rightarrow w_M=0$) $\Rightarrow$ DC well-defined again by construction — what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown below.

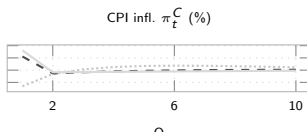
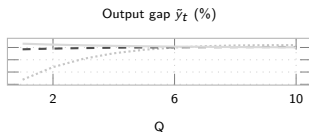
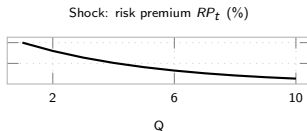
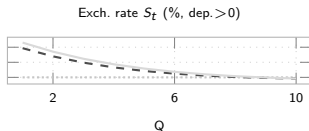


Float: smallest on impact (0.026%) & most persistent; Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

Appendix: Divine Coincidence
(DC) Index Details

IRFs: Foreign Risk-Premium Shock — the Key Mechanism

The shock that decides the ranking: 76% of Peg's welfare loss (Chile calibration). All IRF backup slides in this section are from the earlier aggregate-export specification (not yet regenerated for the sector-specific-export baseline); qualitative shapes and mechanisms are expected to carry over, magnitudes pending re-verification.

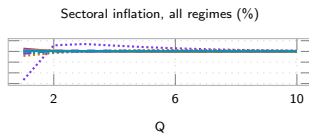
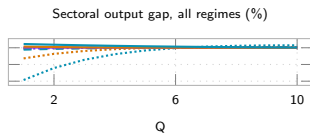


Peg ●●● Managed - - Float —. Slow decay under Float (vs. fast reversal under Peg/Managed) is the persistence mechanism behind Float's larger sectoral dispersion cost. Sector-level IRFs: Appendix (next slide).

Appendix: Additional Shock IRFs
(Backup, not in 25-slide count)

IRFs: Foreign Risk-Premium Shock — Sectoral Detail

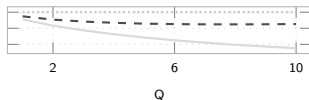
Sector-level counterpart to the previous risk-premium IRF slide — underlies the sectoral dispersion costs on the Network Exposure & Welfare slides.



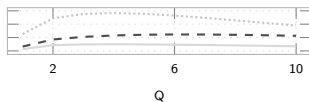
Peg ... Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Resource TFP Shock

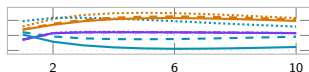
Exch. rate S_t (% dep. > 0)



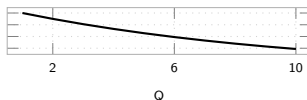
Output gap $\tilde{y}_t$ (%)



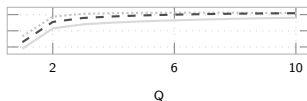
Sectoral output gap, all regimes (%)



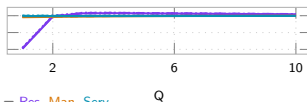
Shock: TFP A_{1t} (%)



CPI infl. π_t^C (%)



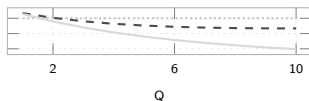
Sectoral inflation, all regimes (%)



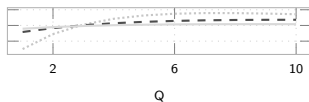
Peg ●●● Managed - - Float Q . Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Manufacturing TFP Shock

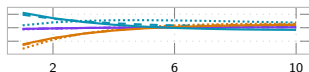
Exch. rate S_t (% , dep.>0)



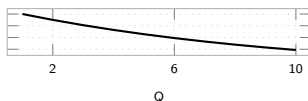
Output gap $\tilde{y}_t$ (%)



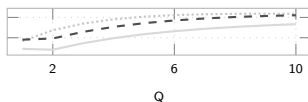
Sectoral output gap, all regimes (%)



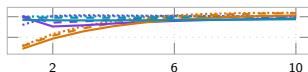
Shock: TFP A_{2t} (%)



CPI infl. π_t^C (%)



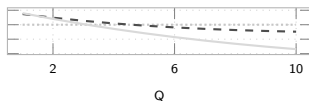
Sectoral inflation, all regimes (%)



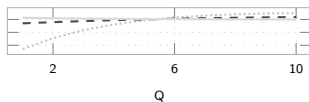
Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv. Q

IRFs: Services TFP Shock

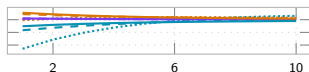
Exch. rate S_t (% , dep.>0)



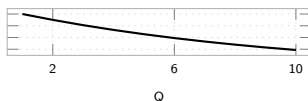
Output gap $\tilde{y}_t$ (%)



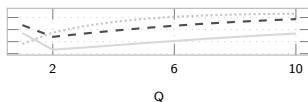
Sectoral output gap, all regimes (%)



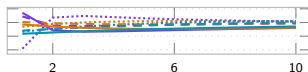
Shock: TFP A_{3t} (%)



CPI infl. π_t^C (%)



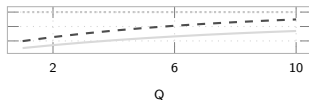
Sectoral inflation, all regimes (%)



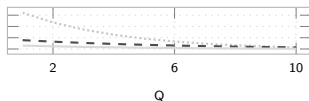
Peg ... Managed - - Float Q . Sectoral by regime: line style = regime, color = Res. Man. Serv. Q

IRFs: Foreign-Demand Shock

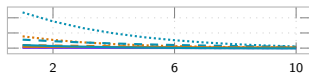
Exch. rate S_t (% dep. > 0)



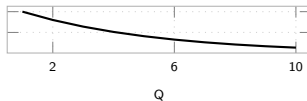
Output gap $\tilde{y}_t$ (%)



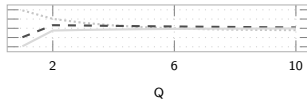
Sectoral output gap, all regimes (%)



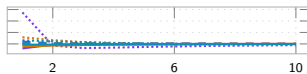
Shock: foreign demand D_t^* (%)



CPI infl. π_t^C (%)



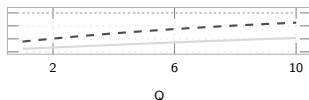
Sectoral inflation, all regimes (%)



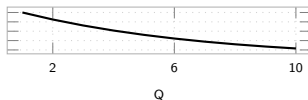
Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Export-Price / ToT Shock

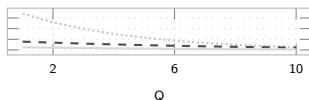
Exch. rate S_t (% , dep.>0)



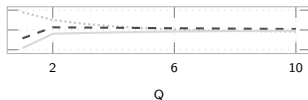
Shock: export price P_t^X (%)



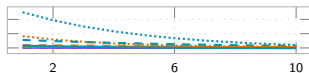
Output gap $\tilde{y}_t$ (%)



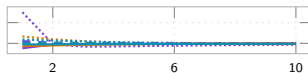
CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)



Sectoral inflation, all regimes (%)

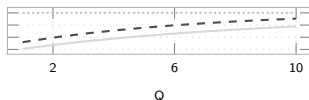


Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

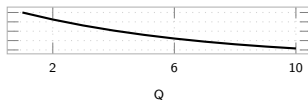
Appendix: Additional Shock IRFs
(Backup, not in 25-slide count)

IRFs: Import-Price Shock (Backup)

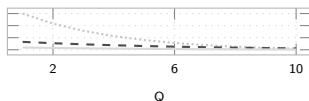
Exch. rate S_t (% , dep. > 0)



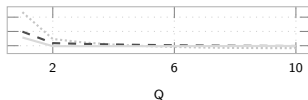
Shock: import price P_t^{F*} (%)



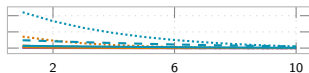
Output gap $\tilde{y}_t$ (%)



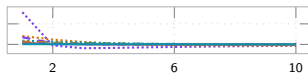
CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)



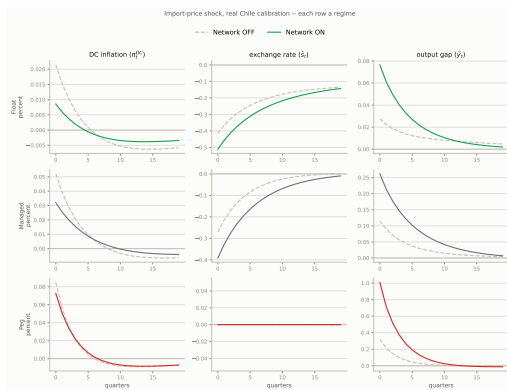
Sectoral inflation, all regimes (%)



Peg ... Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

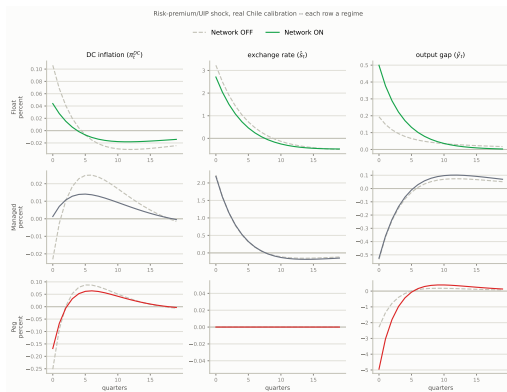
IRFs: Network ON vs. OFF — Import-Price Shock

Full IRFs behind the "Mechanism" slide's table: imported-input price shock, all three regimes, real Chile calibration — **Network OFF** (dashed) vs. **Network ON** (solid):



IRFs: Network ON vs. OFF — Risk-Premium Shock

Full IRFs behind the "Mechanism" slide's table: risk-premium/UIP shock, all three regimes, real Chile calibration — **Network OFF** (dashed) vs. **Network ON** (solid):



Appendix: Additional Shock IRFs
(Backup, not in 25-slide count)

Appendix: Model Equations & Notation (1/2)

Households

$$\max \mathbb{E}_0 \sum_t \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \quad \text{s.t.} \quad P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)] S_t B_{t-1}^*$$

$$C_t^{-\gamma} = \beta(1+i_t)\mathbb{E}_t[C_{t+1}^{-\gamma}/\Pi_{t+1}^C], \quad W_t/P_t^C = C_t^\gamma L_t^\varphi, \quad (1+i_t) = (1+i^*)[1-\psi(\hat{b}_t^* - \bar{b}^*)] \mathbb{E}_t[S_{t+1}/S_t] \text{ (UIP)}$$

C, L : consumption, labor β, γ, φ : discount, RRA, inv. Frisch W, i : wage, dom. rate S, i^* : ER, foreign rate
 ψ : debt-elastic premium B^* : foreign bond P^C : CPI Π_t : profits (rebated)

Firms: Technology and Cost

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F} \Rightarrow MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \quad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \quad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y, A : output, TFP L, X, M : labor, dom./imp. inputs Ω_{ij}^H : dom. IO share Ω_i^F : import cost share
 α_i : labor share MC_{it} : nominal marg. cost P^{F*} : foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Appendix: Model Equations & Notation (2/2)

Firms: Calvo Pricing — reset price (probability δ_i resets, $1-\delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Pi_{i,t+1}^\varepsilon X1_{i,t+1}, \quad X2_{it} = Y_{it} + (1-\delta_i)\beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1}$$

$$P_{it}^\# / P_{it} = \frac{\varepsilon}{\varepsilon-1} X1_{it} / X2_{it}, \quad 1 = (1-\delta_i)\Pi_{it}^{\varepsilon-1} + \delta_i (P_{it}^\# / P_{it})^{1-\varepsilon}, \quad \hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$$

δ_i : Calvo reset prob. (primitive) $\hat{\delta}_i$: derived slope (enters DC weights) $P_{it}^\#$: reset price ε : elasticity (markup $\varepsilon/(\varepsilon-1)$)

Open Economy

$$\hat{b}_t^* = \underbrace{\frac{P_t^{FH}}{\Pi_t^C} = S_t P_t^{F*}}_{\text{LOP}}, \quad \underbrace{(1+i_t) = (1+i^*)[1-\psi(\hat{b}_t^* - \bar{b}^*)]}_{\text{UIP}} \mathbb{E}_t[S_{t+1}/S_t]$$

$$\hat{b}_t^* = \frac{1+i_t-1}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \quad EX_{it} = \kappa_{EX,i} D_t^* \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

$$Y_{it} = \beta_i^H \frac{P_t^H}{P_{it}^H} C_t^H + \frac{P_t^H}{P_{it}^H} EX_{it} + \sum_j \Omega_{ji}^H \frac{MC_{jt} Y_{jt}}{P_{it}^H}$$

$\hat{b}^*$: NFA (scaled) EX_i, IM : sector exports, imports θ^* : export elasticity D^* : foreign demand shifter
 β_i^H : cons. share, sector i $\kappa_{EX,i}$: export-demand scale (baseline: sector-specific) Float: i_t Taylor rule, S_t free Peg: $S_t = \bar{S}$, i_t residual

Appendix: The Peg vs. Float Debate — What Do We Know?

Why it depends (structure):

Mundell (1961) OCA — fix favored by small, open, factor-mobile economies w/ correlated shocks vs. anchor; float favored by large, diversified ones

Frankel (1999) — “no single regime is right for all countries or at all times”; openness, size, trade & capital-market structure all matter

Shock-absorption view (why float wins on real shocks — robustly):

Friedman (1953) — S_t adjusts instead of costly wage/price deflation

Broda (2004) — 75 dev. countries, 1973–96: negative ToT shocks → large real GDP declines under pegs

Edwards & Levy-Yeyati (2005) — 10% ToT deterioration: −0.80pp growth under pegs vs. −0.43pp under floats

Céspedes, Chang & Velasco (2004) — even with dollarized-liability balance-sheet effects, float *still* beats fixed

But floating is not a free lunch (limits on float's insulation):

Calvo & Reinhart (2002) “Fear of Floating” — EMEs peg de facto anyway: credibility-building, not literal welfare optimum

Rey (2013) “Dilemma not trilemma” — under free capital mobility, float alone doesn't deliver monetary autonomy

Consensus: no corner solution is universally optimal — the choice trades a nominal anchor against real-shock absorption, shock- and structure-dependent.

Appendix: The Frontier — Is Fear of Floating Actually Optimal?

Itskhoki & Mukhin (2023) — general open-economy framework with endogenous UIP/PPP deviations: efficient policy = MP closes the output gap + FX intervention eliminates the UIP wedge. Optimal policy is generically a **managed float/crawling peg**, matching observed fear of floating. When the natural RER is stable, this collapses to an “open-economy divine coincidence.”

↪ *vs. us*: **same headline** (Managed, not a corner, is optimal) — reached instead via a multi-sector Calvo network with a persistence-driven, not UIP-gap-closing, mechanism

Bianchi & Coulibaly (2025; 2022 WP) — fear of floating is **optimal**, not behavioral, when credit access is tied to the real exchange rate (self-fulfilling crisis risk). Prescription: float for real shocks, **fix** for financial shocks

↪ *vs. us*: **opposite ranking** for our financial shock — our risk-premium/UIP shock passes straight through under **Peg** (worst regime). Different financial friction (persistent NFA wedge vs. collateral-constraint crisis) flips the prescription

Coulibaly (2023) — discretionary MP is procyclical to offset balance-sheet effects; commitment (inflation targeting) or capital controls raise welfare

↪ *vs. us*: echoes our “Managed = extra instrument” logic — giving policy more tools beats a corner regime

Bottom line: the 2023–25 frontier has also moved past corner solutions — but *why* fear of floating is optimal is mechanism-specific. We add production-network amplification + shock persistence as a distinct reason, not examined by either paper.

Appendix: How Our Results Compare to the Literature

Literature	Says	Us
Mundell (1961); Frankel (1999)	Optimal regime depends on structure, no universal answer	Confirms: ranking flips with shock type (UIP vs. TFP) and network density
Friedman (1953); Broda (2004); Edwards-Levy-Yeyati (2005); Céspedes-Chang-Velasco (2004) Galí & Monacelli (2005)	Float buffers real (ToT/foreign) shocks, lower output loss than peg — robust even to balance-sheet effects Peg dominated, float $\approx$ optimal in 1-sector SOE	Same direction (Peg worst), different shock: risk-premium/UIP, not ToT Same ranking of Peg , but pure Float is not optimal: Managed beats it
Calvo-Reinhart (2002)	EMEs avoid pure float — read as fear , a credibility-driven deviation from optimum	We find partial stabilization is literally welfare-optimal — a persistence-driven, not fear-driven, reason
Rey (2013)	Float alone doesn't insulate; need a second instrument (capital controls/macropu)	Echoes Fanelli-Straub: Managed uses S_t itself as that second instrument
Qiu, Wang, Xu & Zanetti (2026, JME)	Open-economy DC index (CPI+expenditure-switching+profit channels), <i>static</i> , WIOD-calibrated; DC breaks down, output-gap targeting near-optimal	Independent corroboration DC breaks down in open economy; we add the piece they flag as their own top open extension — UIP, NFA persistence, and FX-regime choice

Appendix: The ECB's Open-Economy Production-Network Model

Gnolato, Montes-Galdón & Stamato (2025), “Tariffs across the supply chain,” ECB WP No. 3081 — also underlies Lane’s (ECB Exec. Board) 13 May 2026 speech on energy shocks (4-region extension):

Setup: two-region, multi-sector, full IO structure, international production network; sticky wages/prices; Taylor rule on *domestic* CPI (no regime comparison)

Finding: tariffs on **intermediate** inputs → persistent inflation & larger GDP loss (low substitutability, network propagation); tariffs on **final** goods → one-off inflation spike, smaller GDP loss — *where* in the network a shock hits sets its persistence

↪ *vs. US:* their tariff-location result and our TFP-vs-UIP result are the same insight in two settings: shock **location/type** sets its persistence, echoing our “driver is persistence, not volatility”

What it cites for open economy + production networks:

Network amplification, closed economy: **Pasten, Schoenle & Weber (2020)**; **Ghassibe (2021)**; **Rubbo (2023)**³; **Afrouzi & Bhattarai (2023)**

Open-economy multi-sector NK + networks: **Devereux, Gente & Yu (2023)**; **Hinterlang et al. (2023)** EMuSe (Bundesbank); closest: **Kalemli-Özcan, Soylu & Yildirim (2025)** “Global networks, MP and trade”

Trade fragmentation & inflation: **Moro & Nispi Landi (2024)**; **Ambrosino, Chan & Tenreiro (2026, JIE)**

Optimal MP under tariffs: **Bergin & Corsetti (2023)**; **Bianchi & Coulibaly (2025)** “Optimal MP response to tariffs” (*different* paper from their fear-of-floating one)

Bottom line: none compares FX regimes or asks whether a DC index survives — they fix the monetary rule. That gap is still exactly our question.

³ Published version: Rubbo, E. (2023), *Econometrica* 91(4), 1417–1455 — Production Network Model